# PROGESTERONE

Luteal support & preterm prevention

Drowsiness (oral)

**CHEMICAL NAME** Pregn-4-ene-3,20-dione (natural micronised progesterone)
**CLASS** Natural progestogen (hormone)
**TRADE NAMES** Utrogestan · Crinone (vaginal gel) · Cyclogest · Prometrium

## THERAPEUTIC USES / INDICATIONS

- Prevention of preterm birth in women with short cervix or previous preterm birth (vaginal).
- Luteal phase support in IVF/assisted reproduction.
- Threatened miscarriage with history of miscarriage (vaginal, some guidelines).
- Endometrial protection in HRT (oral); secondary amenorrhoea.

## ADMINISTRATION

- **Vaginal** pessaries, capsules or gel — insert high into vagina; lie down for 15–30 min.
- **Oral** micronised capsules at bedtime (drowsiness).
- **IM** oily injection (deep, painful) in some protocols.

## DOSAGES

- **Preterm birth prevention:** 200 mg vaginally at night from ~16–24 weeks until 34–36 weeks.
- **IVF luteal support:** 200 mg vaginally 3 times daily (Utrogestan) or 8% gel 90 mg once daily (Crinone).
- **Threatened miscarriage:** 400 mg vaginally twice daily.
- **HRT (endometrial protection):** 200 mg orally for 12 days per cycle or 100 mg daily.

## PHARMACOKINETICS

- Vaginal: high uterine levels (first-uterine-pass effect).
- Oral: extensive first-pass metabolism to sedative metabolites.
- **t½:** 5–20 min (plasma) — sustained by formulation.

## PHARMACODYNAMICS

- Binds **progesterone receptors** → secretory transformation of endometrium, maintains pregnancy, ↓ uterine contractility, cervical integrity.
- Metabolites have GABA-A effects → sedation.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Vaginal discharge, irritation.
- Drowsiness, dizziness (oral).
- Breast tenderness, bloating, mood change.

### ADVERSE EFFECTS (SERIOUS)

- VTE (rare).
- Hypersensitivity (IM oily vehicle; peanut/sesame oil in some products).
- Jaundice (rare).

## CONTRAINDICATIONS & CAUTIONS

- Undiagnosed vaginal bleeding; missed miscarriage.
- Breast or genital cancer (hormone-dependent).
- Severe liver disease; active VTE or arterial thrombosis.
- Porphyria; peanut/soya allergy (some products).

### KEY INTERACTIONS

- **Enzyme inducers:** ↓ levels.
- **Ketoconazole:** ↑ levels.

## NURSING MANAGEMENT

- Teach vaginal insertion technique.
- Monitor for bleeding, contractions; cervical length per obstetric plan.
- Oral form: take at bedtime; caution with driving.

## PATIENT EDUCATION

- Insert at night; mild discharge is normal.
- Report heavy bleeding, severe pain or leg swelling.
- Oral capsules may make you sleepy — take at bedtime.

## KEY POINTS

- Natural progesterone.
- Vaginal 200 mg nightly for preterm prevention.
- IVF luteal support.
- Oral causes drowsiness.

**SOURCES CHECKED** **[681]** Utrogestan 100mg Capsules (UK SmPC) **[682]** Progesterone, oral (US label) **[683]** Crinone, vaginal (US label)
*Full references p. 432–457*